



Heavy Metal Stress in Medicinal Plants: Detoxification Mechanisms, Antioxidants, and Implications for Human Health

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Abstract

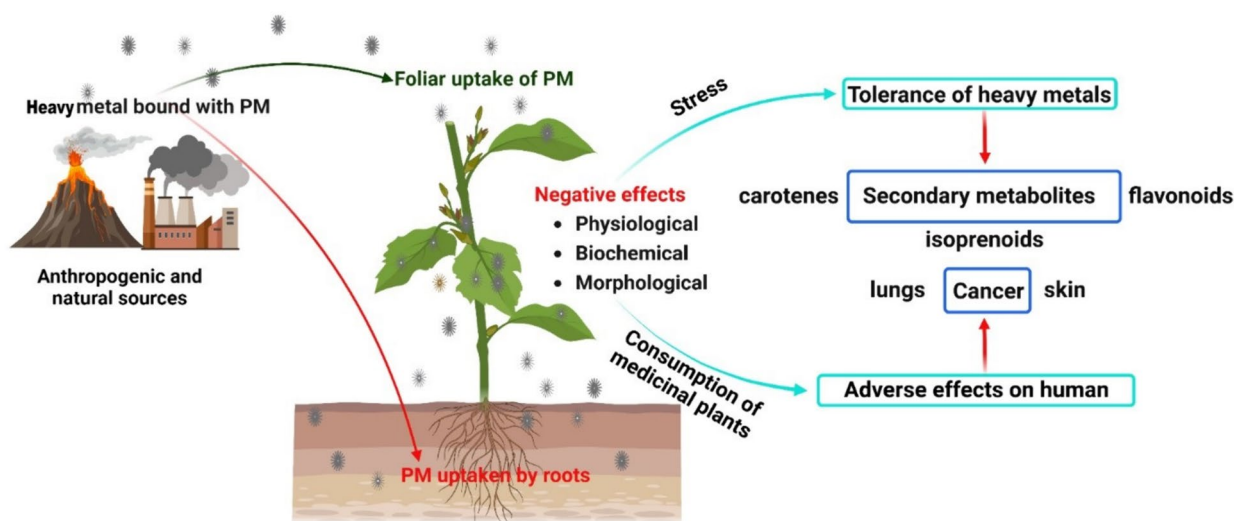
The aromatic and medicinal plants that are used for flavoring, cooking, and medicine make up more than 10% of all vascular plants. Approximately 80% of individuals globally rely on herbal remedies for their medical needs. Heavy metals (HMs) toxicity are mostly caused by anthropogenic activities, although they are also released into the atmosphere and accumulate in soil as a result of natural processes. Metal leaching from landfills, industrial effluent, animal and poultry manure, fertilizer, and pesticide industries, and human activities are among them. Volcanic eruptions and metal smelting are the main sources of numerous HMs. As, Cd, Pb, Cr, and Hg are the most dangerous HMs among those that are released into the environment in terms of their potential toxicity and exposure to living things. HMs accumulation in the environment, such as soil and the atmosphere, is primarily caused by anthropogenic sources. HMs are most effectively transported by particulate matter (PM), which also transports them into plant cuticles through foliar absorption via stomata. It is crucial to detoxify HMs from medicinal plants using both enzymatic and non-enzymatic methods. If HM levels in the plants stay above the recommended threshold during detoxification, it will have an adverse effect on human organs and their functions. A suitable system and procedure should be developed to control the entire process, from the manufacturing of HM-free raw materials to the final consumer good.

Highlights

- About 80% of the world's population uses medicinal plants for primary disease treatment.
- Anthropogenic activity is the primary cause of the deposit and release of heavy metals (HMs).
- HMs are most effectively transported by particulate matter (PM).
- Consumption of HMs accumulated in medicinal plants has adverse effects on humans.
- Regulation of the antioxidant system in plants causes significant detoxification of HMs.

Extended author information available on the last page of the article

Graphical Abstract



Keywords Antioxidant system · Detoxification · Environmental pollution · heavy metals contamination · Human health · Medicinal plants

1 Introduction

Presently, environmental pollution with heavy metals (HMs) is the most critical problem on a global scale, causing damage to plants, animals, and humans via chronic or acute exposure (Saman et al. 2022). Metals with high atomic numbers (> 20) and mass (5 g cm^{-3}) are considered HMs. Based on their function within the biological system, HMs are characterized as either essential or non-essential (Batool et al. 2022). Plants need essential metals such as zinc (Zn), copper (Cu), nickel (Ni), molybdenum (Mo), manganese (Mn), chromium (Cr), and iron (Fe) for their optimum growth and development. Non-essential metals such as lead (Pb), cadmium (Cd), mercury (Hg), and arsenic (As) have no biological role and are toxic to plants even in small quantities (Shahid et al. 2017). Natural and anthropogenic occurrences cause the accumulation and emission of HMs in soil and the atmosphere, although the former is the predominant source of HMs toxicity (Haider et al. 2022). Anthropogenic sources of heavy metal pollution in the environment include metal mining, smelting, fertilizer and pesticide industries, metal leaching from landfills, and industrial effluent (Nachana and Timothy 2019, Chandran 2021). In contrast, volcanic eruptions, metal corrosion, metal evaporation from soil and water, and geological weathering are the main natural sources of environmental HMs contamination. The existence of HMs in the environment may cause a plant to absorb them via its roots and leaves, resulting in metabolic dysfunction in plants, and consumption of these

plants via the food chain causes a significant risk to human health (Shahid et al. 2017).

Medicinal and aromatic plants are used for therapeutic, aromatic, and culinary reasons and account for more than 10% of all vascular plants. Herbal remedies are used to heal almost 80% of the world's population like testicular, lung, and lymphoma cancers can all be successfully treated with podophyllotoxin, which is extracted from *Sinopodophyllum hexandrum* (Royle) Ying, Vinblastine derived from *Catharanthus roseus* (L.) G. Don is an effective treatment for non-Hodgkin's lymphomas, testicular cancer, neck cancer, Hodgkins lymphoma, and pediatric leukemia, Serpentine, a substance that was extracted from *Rauwolfia serpentina* (L.) Benth. ex Kurz (Apocynaceae) roots, is a significant finding for the management of hypertension and lowering blood pressure (Khan and Ahmad 2019). Many studies indicated the uptake of the HMs by medicinal and aromatic plants and their translocation into their edible parts like leaves, shoots, and roots (Tripathi et al. 2021; Sabina et al. 2019; Karahan et al. 2020; Kumar et al. 2021a). HMs contamination alters the chemical composition of plants and can have devastating effects on their efficacy and quality. If medicinal plant species are gathered from industrial areas, there is a high likelihood that the plants will be contaminated with HMs (Sarma et al. 2012). Therefore, caution should be taken when extracting biologically active compounds from plants flourishing on these contaminated lands, as the presence of toxic HMs in plant parts is likely (Anjitha et al. 2021). Numerous secondary metabolites of

plants are utilized to produce pharmaceuticals, fragrances, flavors, dietary supplements, and pigments. Most industrial pharmaceuticals are synthetic modifications or reproductions of naturally occurring plant compounds (Hussain et al. 2012). The most significant bottleneck in producing these compounds is the possibility of overexploitation, as most of the source plants are in the environment. Concerning are the slow growth rate and low concentration of compounds of interest (Atanasov et al. 2015).

Contamination of HMs in medical plants is considered a serious threat to human health (Zhou et al. 2020a, b). Cadmium (Cd) is exceedingly poisonous because it is easily taken up by plant roots and transported to aerial parts. Itai-Itai illness, caused by the deposition of Cd residue in rice (*Oryza sativa* L.), is a well-documented phenomenon (Cai et al. 2020). Lead (Pb), a common HM contaminant with the highest toxicity found in many fertilizers, is easily absorbed and eventually accumulates in the edible sections of medicinal plants (Zhou et al. 2020a, b). It can readily infiltrate the human body through the food chain and endanger people's lives and health. Although the Pb concentration in higher plant parts (such as fruits, seeds, and leaves) is minimal, it remains dangerous when taken by people due to the significant buildup of Pb in the roots (Batool et al. 2022). All of the characteristics above necessitated the development of novel techniques to increase the production of secondary metabolites in medical plants without disturbing their natural populations or allowing HMs ions to enter the final product (Dikilitas et al. 2016, Lajayer et al. 2017).

Plants use morphological, bio-chemical, physiological, and genetic mechanisms to deal with HMs absorbed via foliar and root cells. HMs exposure causes cellular damage in plants by over-production of reactive oxygen species (ROS), which enzymatic and non-enzymatic antioxidants may detoxify (Singhal et al. 2022). Under HMs stress, ROS build and damage cell membranes, increasing membrane lipid peroxidation products (Zhou et al. 2020a, b). Malonaldehyde (MDA) content is a lipid peroxidation biomarker that represents the level of membrane lipid peroxidation and the severity of a plant's response to HMs stress circumstances (Shahid et al. 2017).

The toxicity of HMs has been extensively studied, with numerous investigations focusing on their impact on crop productivity, production of ROS, lipid peroxidation, and potential remediation strategies (Saman et al. 2022; Haider et al. 2022; Batool et al. 2022). However, limited attention has been given to understanding the toxicity of HMs on medicinal plants and how detoxification can overcome their mobilization within plants and food chains. Therefore, the objective of this comprehensive review is to bridge this knowledge gap by providing an in-depth overview of the sources of HMs contamination in the environment, elucidating the toxic effects of HMs, specifically on medicinal

plants, and exploring the regulatory role of the antioxidant system in the effective detoxification of HMs. Additionally, the review will discuss the implications of consuming HM-contaminated medicinal plants on human health. The findings presented in this review underscore the urgent need to implement appropriate mechanisms and systems to ensure the production of raw materials and end-consumer products free from HMs contamination. Such measures are vital for safeguarding human health and preserving environmental quality during medicinal material production.

Although, this study includes, different sources of HMs in the environment, mechanisms involved in the absorption of HMs via foliar and root cells in medicinal plants, classification of medicinal plants based on uptake of HMs, toxicity of HMs in medicinal plants, biosynthesis of secondary metabolites and its functions in HMs stress tolerance, role of nano-biotechnology in the elicitation of secondary metabolites, detoxification role of non-enzymatic and enzymatic antioxidants in medicinal plants and, HMs pose health risks to humans after consuming metal-toxic medicinal plants.

2 Sources of Heavy Metals in the Environment

Human activities (anthropogenic) contribute more than natural processes to the buildup of HMs in the environment. Mines are earthen excavations to extract metals and minerals, and mining is a major source of HM pollution in river basins. The sediments and water samples collected from the headwaters of the Heihe River in Northwest China show that mining activity impacts regional variability and water quality. Heavy metals, including As, Cd, Cr, Cu, Pb, and Zn, were found in those samples (Wei et al. 2018). Besides mining or metal extraction from the earth, HMs are chemically altered at high temperatures to acquire other metals by smelting. It is a method of extractive metallurgy that employs temperature and a chemical reducer to dissolve the ore, removing other materials such as slag and gases and depositing them on the land. Compared to remote regions, the concentrations of HMs were higher close to the smelting companies. Three villages in the Henan province of China surrounded the Pb smelter site, and the concentrations of Cd, Cu, Pb, and Zn rose with decreasing distance to the smelter (Li et al. 2020).

Plant survival depends on plant protection measures; therefore, extensive application of fertilizers and pesticides in agricultural fields enhances HMs in soil (Moameri et al. 2018). Zhou et al. (2020b) investigated the sources of HMs pollution based on published data on Google Scholar and the Web of Science across five decades (1972–2017) and five continents except for Antarctica and Australia. They revealed that fertilizer and pesticides are the primary sources

of twelve HMs (aluminium (Al), As, Cd, cobalt (Co), Cr, Cu, Fe, Hg, Mn, Ni, Pb, and Zn accumulation in the environment. Cu-based pesticides accumulate Cu in soil when applied in vineyards and orchards because Cu is the principal part of the Bordeaux mixture and is used as a fungicide (Rocha et al. 2015). Landfills are excavated pits used to bury pesticide containers and other waste materials that are hazardous to the environment. As a result, HMs contamination in leachate was tested in two Malaysian landfills (Ladang Tanah Merah (LTM) and Sungai Udang (SG)), which are situated in Negeri Sembilan and Melaka, respectively). Quality rating scales (QRS) were employed as a technique for calibration to quantify HMs contamination in leachate. QRS was calculated by dividing HMs concentrations in leachate by the standard-acceptable condition for leachate discharge (EQA 2009). As and Cr exhibited 787 and 552 QRS values, respectively, suggesting increasing degradation of landfill leachate contamination. The leachate is heavily contaminated if QRS values surpass 300 percent (Hussain et al. 2021).

Global industrial wastewater's application has dramatically increased to irrigate agricultural land to address water shortages (Hussain et al. 2019a; Zwolak et al. 2019). Therefore, roughly 20 million hectares of fertile land are irrigated with wastewater containing HMs, depositing HMs into the soil for long-term usage (Khalid et al. 2018). HMs such as As, Cu, Cr, Cd, Ni, Zn, Pb, and Hg primarily derive from industrial waste, damaging marine and terrestrial environments. Their bio-accumulative toxicity in biotic systems results from their high solubility in aquatic settings (Shrestha et al. 2021). In the event of water scarcity, wastewater is utilized as an alternative. As a result, organically generated fertilizer is employed in fields to compensate for synthetic fertilizer shortages. Furthermore, animal dung often supplies secondary micronutrients typically lacking in commercial fertilizers. Cattle contributed the most nitrogen (N) to global manure production (43.7 percent), while poultry, goats, and sheep generated one-third of the total manure N production globally (Zhang et al. 2014). Therefore, the long-term application of excessive manure on land also accumulates HMs in the soil. For example, adding pig dung to the soil for 16 years increased its Cd, Cu, and Zn content and also increased its productivity (Wang et al. 2020).

Apart from anthropogenic activities, natural phenomena are also responsible for HMs buildup in the soil. Therefore, combined plasma mass and atomic fluorescence spectrometry of agricultural soil in Guangdong Province of China demonstrated the presence of As, Cu, Cr, and Ni, which are accumulated predominantly from natural sources (Cai et al. 2019). Volcanic activity is proof of the Earth's core's dynamic nature and a manifestation of a short-lived intensive crustal movement that created ash and pyroclastic debris that persisted in the environment for days to millennia. These pyroclastics

and volcanic ash include HMs such as As, Hg, etc. (Kapoor and Singh 2021; Ermolin et al. 2018). The corrosion of HMs causes them to become oxidized and exposed to the atmosphere. The Colombo Metropolitan Region (CMR), Sri Lanka's capital, is an industrialized metropolis with a high volume of motor vehicles. Cu was detected in high amounts north of the Colombo port, where vast quantities of metallurgical components are housed. High corrosion of these metallic components caused by contact with ocean water enables the incorporation of Cu into the surface soil (Herath et al. 2018). In addition to corrosion, evaporation also contributes to incorporating HMs into the top-soil surface. Therefore, a Saudi Arabia (SA) laboratory investigation indicated the upward migration of metals in soils when evaporation is the predominant transport mechanism. Above the layer of loamy soil, sand is the dominant soil type in the dry area of South Africa, and HMs are kept in loamy soil (Lima et al. 2014). At the same time, evaporation drives HMs to the surface of sandy soil, causing them to collect on the soil surface (Lima et al. 2014). The weathering of rocks is regarded as the most prominent contributor to the accumulation of HMs in the environment. The concentration and composition of HMs are strongly controlled by the nature of the rock and the surrounding environment (Srivastava et al. 2017). The lithogenic sources, such as andesite and basalt rocks, are rich in Cr and Ni, respectively, and their weathering may lead to these HMs accumulating on the soil's surface (Ayala-Perez et al. 2021). Thus, it has been shown that human and natural occurrences are responsible for the buildup of HMs in the environment. Table 1 displays the deposition of HMs in the soil and water due to human activities and natural occurrences.

3 Heavy Metal Stress in Medicinal Plants

3.1 Importance of Medicinal Plants

Using medicinal plants has been a longstanding practice within human populations, with ancient wisdom passed down through generations and shared among communities (Ahmad et al. 2017). Numerous ethnobotanical studies have been conducted in various countries worldwide to investigate the trade of herbal medicines. These studies include research conducted in Syria, Israel, Iran, Jordan, South Africa, Greece, Ethiopia, Peru, Turkey, Botswana, Nepal, Pakistan, India, and China (Ahmad et al. 2017). Having been used as traditional folk treatments for millennia, medicinal plants now play a significant part in the pharmaceutical, health food, and natural cosmetics sectors. In China, herbal medication cured 85 percent of COVID-19 patients with positive test findings from all infection phases, some examples of these are *Citrus Spp.*, *Camellia sinensis* O Ktunze [Thea sinensis L.], *Allium sativum* L.,

Table 1 Heavy metal concentration in the environment due to different sources

Heavy metal	Country	Medium	Event type	Sources	Concentration level (Soil-mg kg ⁻¹) and (Water-mg L ⁻¹)	References
Antimony (Sb)	China	Soil	Anthropogenic	Steel industry	3.39	(Yan et al. 2020)
				Petrochemical industry	2.88	
				Garbage incineration plant	2.62	
	France	Soil	Anthropogenic	Sb processing plant	1654.46	(Zhou et al. 2019)
				Mining site	691–1986	(Elbaz-Poulichet et al. 2020)
				Roadside soil	1.75	(Thestorf and Makki 2021)
				The Mau Due mine	47–95	(Cappuyns et al. 2021)
Arsenic (As)	China	Soil	Anthropogenic	Fuel combustion	41.3	(Wu et al. 2022)
				Sb mining smelting process	147.83–215.91	(Ren et al. 2018)
				Agnano effluent	Upper layer sediment = 137, lower layer sediment 205	(Buccino et al. 2021)
	Italy	Marine sediment	Semi-anthropogenic (halfway between geogenic and anthropogenic)			
	India	Water	Anthropogenic	Fertilizer usage	0.05	(Rahman et al. 2021)
				Coal combustion and agricultural activities	6.25	(Kumar et al. 2021b)
Beryllium (Be)	Kenya	Water	Anthropogenic	Agricultural and urban runoff	0.008	(Githaiga et al. 2021)
				Metal smelting	46–209	(Xing et al. 2020)
	China	Soil	Anthropogenic	Fuel combustion	31	(Wu et al. 2022)
	Bangladesh	Soil	Anthropogenic	Coal combustion and agricultural activities	3.04	(Kumar et al. 2021b)
Cadmium (Cd)	Bangladesh	Soil	Anthropogenic	Coal combustion and agricultural activities	0.25	(Kumar et al. 2021b)
	China	Soil	Anthropogenic	Agriculture activities, human population density, number of factories, and distance of sampling sites to major road	0.13–1.08	(Li et al. 2021)
Chromium (Cr)	China	Soil/ water	Anthropogenic	Agriculture farm, household waste, sewage outlet, paper factory, glass industry, textile mill, paper factory, printing, and dyeing plant	0.09–1.08	(Yin et al. 2021)
	Kenya	Water	Anthropogenic	Agricultural and urban runoff	0.0004	(Githaiga et al. 2021)
				Metal smelting	16–180	(Xing et al. 2020)
	Bangladesh	Soil	Anthropogenic	Coal combustion and agricultural activities	56.8	(Kumar et al. 2021b)

Table 1 (continued)

Heavy metal	Country	Medium	Event type	Sources	Concentration level (Soil-mg kg ⁻¹) and (Water-mg L ⁻¹)	References
Copper (Cu)	China	Soil/ water	Anthropogenic	Agriculture farm, household waste, sewage outlet, paper factory, glass industry, textile mill, paper factory, printings, and dyeing plant	41.36–562.62	(Yin et al. 2021)
	China	Marine sediment	Anthropogenic and natural	Agricultural and domestic sewage and weathering	58.5–81.8	(Yang et al. 2021)
	Kenya	Water	Anthropogenic	Agricultural and urban runoff	0.002	(Githaiga et al. 2021)
	China	Soil	Anthropogenic	Smelting slag	1.9	(Liu et al. 2018)
	Bangladesh	Soil	Anthropogenic	Coal combustion and agricultural activities	20.6	(Kumar et al. 2021b)
	China	Soil/ water	Anthropogenic	Agriculture farm, household waste, sewage outlet, paper factory, glass industry, textile mill, paper factory, printing, and dyeing plant	11.35–109–15	(Yin et al. 2021)
Cobalt (Co)	China	Marine sediment	Anthropogenic and natural	Agricultural and domestic sewage and weathering	15.1–37	(Yang et al. 2021)
	China	Soil	Natural	Weathering, soil parent material	39.7	(Wu et al. 2022)
	Bangladesh	Soil	Anthropogenic	Coal combustion and agricultural activities	11.45	(Kumar et al. 2021b)
	China	Soil/ water	Anthropogenic	Agriculture farm, household waste, sewage outlet, paper factory, glass industry, textile mill, paper factory, printing, and dyeing plant	3.80–43.04	(Yin et al. 2021)
Iron (Fe)	Kenya	Water	Anthropogenic	Agricultural and urban runoff	0.0007	(Githaiga et al. 2021)
	Bangladesh	Soil	Anthropogenic	Coal combustion and agricultural activities	28159.5	(Kumar et al. 2021b)
Lead (Pb)	Bangladesh	Soil	Anthropogenic	Coal combustion and agricultural activities	22.35	(Kumar et al. 2021b)
	China	Soil/ water	Anthropogenic	Agriculture farm, household waste, sewage outlet, paper factory, glass industry, textile mill, paper factory, printing, and dyeing plant	4.76–107.69	(Yin et al. 2021)
	China	Marine sediment	Anthropogenic and natural	Agricultural and domestic sewage and weathering	18–36.8	(Yang et al. 2021)
	Kenya	Water	Anthropogenic	Agricultural and urban runoff	0.004	(Githaiga et al. 2021)

Table 1 (continued)

Heavy metal	Country	Medium	Event type	Sources	Concentration level (Soil-mg kg ⁻¹) and (Water-mg L ⁻¹)	References
Manganese (Mn)	China	Soil	Anthropogenic	Metal smelting	1280–5440	(Xing et al. 2020)
	Bangladesh	Soil	Anthropogenic	Coal combustion and agricultural activities	448.5	(Kumar et al. 2021b)
Mercury (Hg)	China	Marine Sediment	Anthropogenic	Seepage	568	(Lin et al. 2021)
	Kenya	Water	Anthropogenic	Agricultural and urban runoff	0.058	(Githaiga et al. 2021)
	European Union	Soil	Anthropogenic	Mining and coal combustion	.051	(Panagos et al. 2021)
	Portugal	Marine Sediments	Anthropogenic	Industrial waste	1.2	(Vieira et al. 2021)
	Kenya	Water	Anthropogenic	Agricultural and urban runoff	0.0008	(Githaiga et al. 2021)
	Korea	Soil	Anthropogenic	Smelting industry	0.21–17	(Jeong et al. 2021)
Molybdenum (Mo)	Bangladesh	Soil	Anthropogenic	Coal combustion and agricultural activities	0.3	(Kumar et al. 2021b)
	Kenya	Water	Anthropogenic	Agricultural and urban runoff	0.066	(Githaiga et al. 2021)
Nickel (Ni)	China	Marine Sediment	Anthropogenic	Seepage	0.4–29	(Lin et al. 2021)
	Bangladesh	Soil	Anthropogenic	Coal combustion and agricultural activities	25.05	(Kumar et al. 2021b)
	China	Soil/ water	Anthropogenic	Agriculture farm, household waste, sewage outlet, paper factory, glass industry, textile mill, paper factory, printing, and dyeing plant	10.39–78.50	(Yin et al. 2021)
Selenium (Se)	Kenya	Water	Anthropogenic	Agricultural and urban runoff	0.162	(Githaiga et al. 2021)
	Kenya	Water	Anthropogenic	Agricultural and urban runoff	0.006	(Githaiga et al. 2021)
	Bangladesh	Soil	Anthropogenic	Coal combustion and agricultural activities	4.18	(Kumar et al. 2021b)
Titanium (Ti)	Bangladesh	Soil	Anthropogenic	Coal combustion and agricultural activities	4011	(Kumar et al. 2021b)
Uranium (U)	Bangladesh	Soil	Anthropogenic	Coal combustion and agricultural activities	3.5	(Kumar et al. 2021b)
Vanadium (V)	China	Marine Sediment	Anthropogenic	Seepage	1.9–16	(Lin et al. 2021)
	China	Soil	Natural	Weathering, soil parent material	348	(Wu et al. 2022)
	Bangladesh	Soil	Anthropogenic	Coal combustion and agricultural activities	71.3	(Kumar et al. 2021b)
Zinc (Zn)	Bangladesh	Soil	Anthropogenic	Coal combustion and agricultural activities	58.5	(Kumar et al. 2021b)

Table 1 (continued)

Heavy metal	Country	Medium	Event type	Sources	Concentration level (Soil-mg kg ⁻¹) and (Water-mg L ⁻¹)	References
	China	Soil/ water	Anthropogenic	Agriculture farm, household waste, sewage outlet, paper factory, glass industry, textile mill, paper factory, printing, and dyeing plant	40.60–502.21	(Yin et al. 2021)
	China	Marine sediment	Anthropogenic and natural	Agricultural and domestic sewage and weathering	51.5–92.7	(Yang et al. 2021)
	Kenya	Water	Anthropogenic	Agricultural and urban runoff	0.007	(Githaiga et al. 2021)

mg kg⁻¹ Milligrams per kilogram, mg L⁻¹, Milligrams per litre

Allium cepa L., *Mentha × piperita* L., and *Nigella sativa* L. are a few antiviral medicinal plant species (Demeke et al. 2021). According to the World Health Organization (WHO), around sixty percent of the world's population and 80% of the developed world's population are healed of their primary diseases by plant-based medications. The Amazon rainforest, Andean highlands, and tropical and subtropical woods of Central America make Latin America the most diverse location for medicinal plants. In general, rich species diversity increases the use of medicinal plants, and 25,000 plants were examined for their therapeutic efficacy worldwide. A thorough investigation in the Amhara Regional State of Ethiopia's Enemay District revealed 92 plants, belonging to 84 genera and 48 botanical families, with noteworthy medicinal qualities for treating human and animal illnesses (Birhan et al. 2023). In addition to their therapeutic value, medicinal plants accumulate HMs through foliar and root absorption (Ranilla et al. 2010; Calixto 2005). As was detected in agricultural soil and stream water in each of the twenty nations of Latin America, all are under acute heavy metal stress (Bundschuh et al. 2021). Van (2015) reviewed Africa's economically significant medicinal plants and determined that over 5400 plant species have historically been used as medicines. Eighty-five of these species are extensively traded internationally as nutritional supplements for human use. *Agathosma betulina* L., *Boswellia sacra* L., *Boswellia papyrifera* L., *Cyclopia genistoides* L., *Hibiscus sabdariffa* L., *Hypoxis hemerocallidea* L., *Hoodia gordonii* L., *Lessertia frutescens* L., *Mondia whitei* L., *Pelargonium sidoides* L., *Sceletium tortuosum* L., and *Tylosema esculentum* L. are the most popular species of medicinal plants found in Africa. *Anoectochilus roxburghii* L., the most frequently grown Chinese herb, is a tonic for patients with cough-variant asthma, chronic hepatitis B, type 2 diabetes, hyperuricemia, etc. China is the world's largest manufacturer of medicines derived from herbs to treat various disorders (Ye et al. 2017). Similarly, in Pakistan, various medicinal plants, i.e., *Acacia concinna* L., *Colchicum luteum* L., *Holarrhena pubescens* L., *Narthex asafetida* L., *Quercus oblongata* L., *Senna alexandrina* L., *Saussurea heteromalla* L., and *Ziziphus nummularia* L. have been identified as having unique traditional applications that show promise in the treatment of various diseases (Ahmad et al. 2018). These plants exhibit the highest ethnobotanical index values, indicating their potential utilization in herbal practices across different healthcare systems.

3.2 Particulate Matter as a Carrier of Heavy Metals for Their Translocation in Medicinal Plants

Particulate matter (PM) is the most significant carrier of HMs and translocates them into plant cuticles through foliar absorption via stomata (Gao et al. 2022). PM is a mixture

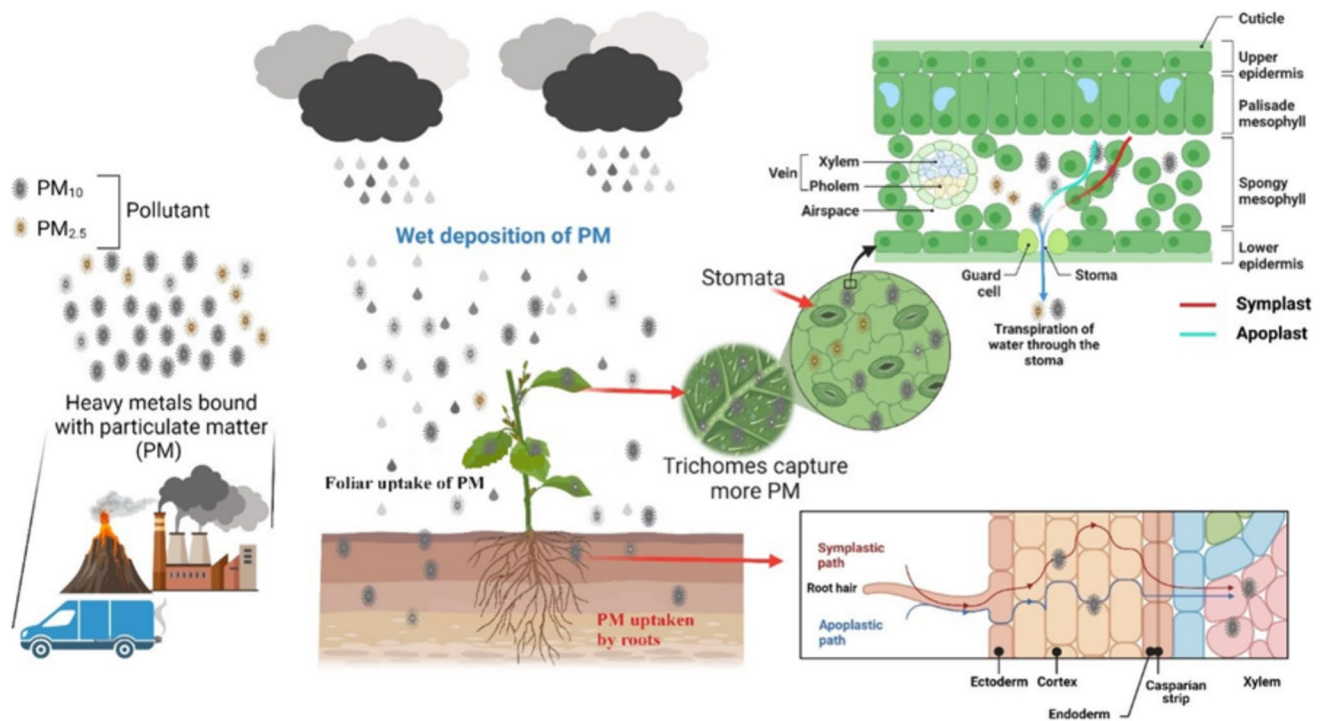


Fig. 1 Uptake of heavy metal bounded particulate matter in medicinal plants from different sources. PM (Particulate Matter), a mixture of heterogeneous components from anthropogenic and natural sources,

is a rich carrier of essential and non-essential HMs (Heavy Metals) enters into plants through leaves and roots

of heterogeneous components from anthropogenic and natural sources and a rich carrier of essential and non-essential HMs such as Zn, Pb, and Ni (Sharma et al. 2020; Potter et al. 2021). According to the WHO, PM₁₀ having a diameter of 10 µm or less is considered a pollutant (WHO 2021). The surface of PM adsorbs the toxic HMs, and the leaves' adaxial and abaxial surfaces are the primary receptor sites for PM deposition. The leaves' epicuticular wax layer and trichomes trapped and retained the PM with HMs. Chen et al. (2017) discovered that the quantity of PM_{2.5} captured by the foliar surface is positively linked with the leaf's total trichomes. After the deposition of PM on the foliar surface, the leaves' surface became injured due to the alkaline nature of PM. It enhanced the permeability of metals from the cuticular pores and stomata. Stomatal openings in the leaf cuticle are the main pathways for penetrating HMs that have leached out from the PM, crossed the cuticular barrier, and reached intercellular and cellular spaces (Fig. 1). Generally, stomata facilitate the penetration of particles of 43 nm, whereas hydrophilic substances ranging from 0.6–4.8 nm are penetrated through cuticular aqueous pores, thus restricting the transport of particles larger than 4.8 nm (Shabnam et al. 2021). The translocation of HMs through PM depends upon the following factors: leaf morphology, HM speciation, meteorological conditions, plant development stage,

and physiology (Shahid 2017). Lettuce (*Lactuca sativa* L.), a member of the *Asteraceae* family, has been used traditionally due to its many therapeutic properties. It is very sensitive regarding HMs toxicity and uptake of metals through foliar absorption (Kolahi et al. 2020). PM containing cadmium oxide (CdO) and lead oxide (PbO) were released into the atmosphere from human sources such as coal combustion and Pb recycling industries and were absorbed by stomatal openings of lettuce leaves. The bio-accessibility of Cd is greater than Pb because Cd has higher solubility and a smaller size than Pb, making it preferable to enter through stomata (Xiong et al. 2019).

3.3 Mechanism of Root Uptake of HMs in Medicinal Plants

With the increasing demand for minerals, the extraction of metalliferous mining has been increasing in recent centuries, particularly for Pb, Ag, and Cd, which ultimately enhanced the toxic HMs in soil (Moreno et al. 2002). These toxic HMs are taken up by medicinal plants through roots by binding to polysaccharides on the rhizodermal cell surface. After root adsorption, HMs penetrate passively and diffuse through translocating water streams. HM uptake depends upon plant or soil type, metal speciation in soil, soil particle

Table 2 Level of non-essential Heavy Metals in different types of medicinal plants

Plant species	Types of medicinal plant	Heavy metal	Enrichment factor (EF)	Translocation factor (TF)	Country	Sampling size	Sample medium	Experimental status	Effects	References
<i>Aster tataricus</i> <i>Lf.</i> , <i>Salvia miltiorrhiza</i> <i>Bge.</i> , <i>Radix Aucklandiae</i> , <i>Scutellaria baicalensis</i> <i>Georgi</i>		Cr, Ni, Cu, Zn, As, Cd, Hg and Pb			China	7 sampling sites	Soil and different parts of plants	Laboratory	The presence of heavy metal pollution in therapeutic herbs and native soil do not strongly correlate	(Meng et al. 2022)
<i>Ocimum gratissimum</i> , <i>Vernonia amygdalina</i> , <i>Spinacia oleracea</i> and <i>Citrus aurantifolia</i>	High accumulation	Pb, Mn, Zn, Cr, Cu, Fe	0.5, 1, 1.4, 3.2	0.6, 0.2, 0.6, 0.7	Nigeria	40	Leaves	Laboratory	The HMs % varied in the same medicinal plant from several areas, showing the effects of anthropogenic sources associated with oil	(Obi Iyeke 2021)
<i>Typha latifolia</i>	Low/high accumulating	As, Cd, Cr, Pb	0.5, 1, 1.4, 3.2	0.6, 0.2, 0.6, 0.7	Iran	30	Soil and plant parts	Laboratory	This species accumulates the Pb significantly greater than As, Cd and Cr	(Haghnazar et al. 2021)
<i>Avicennia marina</i>	Low/high accumulating	As, Mo, Li, Pb, Sb, Ti	More than 1	0.6, 0.7, 0.3, 0.8, 0.6, 0.7	India	36	Soil and plant parts	Laboratory	Non-essential HMs are high EF but low TF to shoot, Sb shows highly extremely EF (> 40) in both pre and post-monsoon season	(Maurya and Kumari 2021)
<i>Ocimum basilicum</i>	Low/high accumulating	Cd, Co, Cr, Pb	3, 1.14, 2, 5	0.10–0.79	Romania	NI	Soil and plant parts	Pot experiment	Maximum EF was found for Cd and Pb and these four HMs are not significantly translocated to shoots	(Dinu et al. 2020)

Table 2 (continued)

Plant species	Types of medicinal plant	Heavy metal	Enrichment factor (EF)	Translocation factor (TF)	Country	Sampling size	Sample medium	Experimental status	Effects	References
<i>Athyrium esculentum</i> , <i>lentum</i> ,	Low/high accumulating	Cd, Pb	1.76, 1.49	0.25, 0.1	Malaysia	NI	Soil and plant parts	Laboratory	<i>A. esculentum</i> showed high accumulation of Cd and Pb from soil to roots and low accumulation to shoots	(Sulaiman and Hamzah 2018)
<i>Chromolaena odorata</i>	Low/high accumulating	Cd, Pb	1.50, 0.80	0.1, 0.3	Malaysia	NI	Soil and plant parts	Laboratory	<i>C. odorata</i> showed high EF for Cd and Pb but low TF from roots to shoots	(Sulaiman and Hamzah 2018)
<i>Lantana camara</i>	Low/high accumulating	Cd, Pb	2.59, 1.74	0.9, 0.7	Malaysia	NI	Soil and plant parts	Laboratory	<i>L. camara</i> showed high EF of Cd and Pb from soil to roots and low EF to shoots	(Sulaiman and Hamzah 2018)
<i>Aloe barbadensis</i>	Low/high accumulating	As, Sb	19.38–28, 18.01–29.12	0.29–0.67, 0.51–0.89	Malaysia	NI	Soil and plant parts	Pot experiment	<i>A. barbadensis</i> deposit higher amount of As and Sb in plant roots from soil but TF is low to shoot	(Ong et al. 2018)
<i>Mentha piperita</i>	Low/high accumulating	Cr, Mo,	More than 1	Lower than 1	Romania	NI	Soil and plant parts	Pot experiment	EF in <i>M. piperita</i> is high for Cr and Mo into roots from soil and TF is low towards shoots	(Dinu et al. 2021b)

Table 2 (continued)

Plant species	Types of medicinal plant	Heavy metal	Enrichment factor (EF)	Translocation factor (TF)	Country	Sampling size	Sample medium	Experimental status	Effects	References
<i>Saponaria officinalis</i>	High accumulating	As, Cd	128.8, 57.9	More than 1	Serbia	7	Soil and plant parts	Laboratory	In <i>S. officinalis</i> significant EF and TF for As and Cd from soils to roots and then to shoots	(Nujkic et al. 2020)
<i>Achillea millefolium</i>	High accumulating	As, Cd	566.3, 306.2	More than 1	Serbia	7	Soil and plant parts	Laboratory	<i>A. millefolium</i> have significantly high EF and TF of As and Cd from soils to roots and then to shoots	(Nujkic et al. 2020)
<i>Pistia stratiotes</i>	High accumulating	Cd, Cr, Pb	1.01–3.37	More than 1	India	NI	Sugar mill effluent and plant parts	Field and Laboratory	High EF and TF of HMs indicated that high mobility of these from soil to roots and then to shoots and leaves	(Kumar et al. 2018)

As Arsenic, Cd Cadmium, Pb Lead, Hg Mercury, Cr Chromium, Cu Copper, Co Cobalt, Fe Iron, Zn Zinc, Br Bromine, Th Thorium, Ni Nickel, Hg Mercury, Mn Manganese, Mo Molybdenum, Li Lithium, Sb Antimony, Ti Titanium, HMs Heavy Metals

size, cation-exchange capacity, soil pH, organic content, and microbial activity (Li et al. 2022). Different transporters help move metal from roots to aerial parts of medicinal plants through xylem loading via symplast or apoplast pathways across the root cortex to plant tissues (Fig. 1). Metals form complexes with phytochelatin while passing through xylem sap (Pourrut et al. 2011). Phyto-chelators are included for the transportation of metals governed by transpiration. Different transporter gene families, including ATP-binding cassette (ABC) transporters, are the largest and oldest transporter families that facilitate the transportation of HMs to aerial parts of the plant. *LsPCS1* and *LsPCS2* are the two primary transporter genes that transport Cd to the aerial tissues of parts of lettuce (Yazdi et al. 2019).

3.4 Classification of Medicinal Plants Based on Uptake of HMs

Globally, the contamination of ecosystems by HMs is a matter of great concern. Approximately 19.4% of China's arable land is subject to metal stress (Zhao et al. 2015). HMs are hazardous to plants and humans due to their low mobility, slow migration from soil to underground water, and accessible availability in aerial plant parts. Using green plants to remove, confine, or lessen dangerous environmental pollutants is a cheap, safe, and environmentally acceptable remediation technique known as phytoremediation (Awasthi et al. 2024). According to Rodrigues et al. (2011), Pb migrates slowly from soil to subterranean water, where it becomes readily available for plants. As a result, the Pb concentration of aerial plant components like leaves, fruits, and seeds is higher. People have exhibited a growing interest in natural sources of substances and herbal medicines for therapeutic reasons during the last century, and according to the WHO, nearly 4 billion people use plant-based medicines for primary health care. Even though HMs adversely affect the effectiveness and quality of herbal remedies, excessive HMs in traditional Chinese medicine (TCM) have sometimes tarnished the industry's reputation by generating economic losses (Zhao 2014). To combat the accumulation of HMs in the environment, phytoremediation is the superior or more effective method in which plants absorb HMs effectively and exhibit fast development, such as hyper-accumulating plants. In contrast, the biomass output of low-accumulating plants is greater despite their lower absorption capacity than that of high-accumulating plants. Estimating a plant's relationship to the soil for HMs accumulation in shoots and roots is measured by enrichment factors (EF) and translocation factors (TF). The medicinal plants are considered low-accumulating when EF and TF are less than 1 and vice versa (Zhou et al. 2020a). Different medicinal plants

were identified based on their EF and TF of HMs are presented in Table 2.

3.5 Toxicity of HMs in Medicinal Plants

Globally, HMs toxicity is the main danger from which plants are destroyed. The release of HMs into the environment by many natural and human-made sources has detrimental consequences for plants. The toxicity of HMs to plants depends on how these metals enter the plant, i.e., via the roots or the leaves (Shahid et al. 2017). Depending on which plant organ is directly exposed to HMs, the exact applied quantities of metals may have varying effects on the plant. Yang et al. (2005) examined the combined application of Cd and Zn to sedum (*Sedum alfredii* L.) via roots. They revealed that the accumulation of HMs depends on the varying concentration of Cd and Zn. The uptake of HMs through roots induces reduced germination of seeds, root elongation, and plant biomass and inhibits chlorophyll biosynthesis. Moreover, plant cytology and physiology, such as enzymatic reactions, nutrient balance, photosynthesis, and respiration, are also disturbed by accumulation of HMs in medicinal plants (Kiran et al. 2022). Although foliar uptake of HMs by medicinal plants also induces toxic effects on plants' different metabolic and physiological processes. However, more research is required to determine the level of toxicity caused by HMs foliar absorption (Schreck et al. 2012).

The ROS are the byproducts of naturally occurring aerobic reactions in living organisms. Depending on their abundance in plants, organelles such as chloroplasts, mitochondria, and peroxisomes that create ROS are highly reactive, unstable, short-lived, and serve a dual function as detrimental and helpful species. The ROS are comprised of alkoxyl ($\text{RO}\bullet$), hydrogen peroxide (H_2O_2), hydroxyl ($\text{HO}\bullet$), peroxy ($\text{RO}_2\bullet$), single oxygen ($1/2\text{O}_2$), superoxide anion ($\text{O}_2\bullet^-$), and hydroperoxide radicals (ROOH). Due to the toxicity of HMs, plant tissues produce more ROS, which is considered a negative stress-induced component (Pourrut et al. 2013). This imbalance in ROS generation in medicinal plants results in physiological alterations such as oxidative stress, which eventually leads to cell death through membrane lipid peroxidation, DNA and RNA damage, enzyme inhibition, and protein oxidation (Van et al. 2018; Martinez et al. 2017; Li et al. 2018; Maleki et al. 2017). Alone and combined application of Cd with Ni as a foliar treatment for White Air plant (*Tillandsia albida* L.) significantly increased H_2O_2 generation, which causes oxidative stress (Kovacik et al. 2012). HMs cause oxidative stress in plants through both root uptake and foliar absorption. For mustard (*Brassica juncea* L.), the growth medium containing CdCl_2 also increased the generation of ROS, especially H_2O_2 , which eventually resulted in oxidative stress (Kapoor et al. 2019a). In addition to physiological activities, HMs cause

Table 3 Evidence of the negative effect of HMs on physiological, biochemical, and morphological features of medicinal plants

Plant species	Heavy metal	Substrate	Treatment	Exposure duration	Effects	References
<i>Echium amoenum</i>	Se	Foliar spray	2, 4, 8 and 16 mg L ⁻¹	2 weeks before flowering	A 4 mg L ⁻¹ sodium selenite (SS) increased flower yield. At the end of flowering, the 8 mg L ⁻¹ SS had increased total phenol and flavonoid concentrations, antioxidant activity, soluble sugars, root, and fresh weight. At the height of blooming, 4 mg L ⁻¹ SS showed greater total chlorophyll levels. The maximum amount of Se acclimation was released in the petals at 16 mg L ⁻¹ sodium selenite. Twenty composites with α -Phene (23.61%) with SS (4 mg L ⁻¹) were found	(Kalaie et al. 2022)
<i>Aloe ferox</i>	Cd	Solution	50 mmol L ⁻¹	Once a week throughout the cultivation	Cd had an impact on the morphology of the leaves, reduced biomass, changes abaxial and adaxial epidermis of leaves, changed the way cuticles were laid down and increased the number of stomata. Roots treated with Cd had reduced levels of lignification and peroxidation of membrane lipids, high in Cd concentration. Cd moving to the leaves could be harmful because active secondary metabolites that are primarily found in the leaves	(Sirova and Vaculik 2022)

Table 3 (continued)

Plant species	Heavy metal	Substrate	Treatment	Exposure duration	Effects	References
<i>Andropogonis paniculata</i>	Zn, Cr, Cu	Hogland nutrient solution	25–175ppm	10 days	A 50 ppm Cu treatment results in a reduction in carbohydrate content whereas a 25 ppm Cu treatment results in a very little rise in carbohydrate content relative to control. Even at 25 ppm, Zn and Cr produce a decrease in carbohydrate content. Cu and Zn concentrations more than 125 ppm result in a decrease in protein content. Protein content in the plants treated with Cr has decreased from a value of 75 ppm	(Hridhya and Anitha 2022)
<i>Citrullus colocynthis</i>	Ni	Nutrient solution	50–200 uM NiCl ₂	Daily till to germination	Increased stress results in a drop in germination %, root length, shoot length, fresh weight, dry weight, and the number of leaves while an increase in Na ⁺ , Cl ⁻ , Ni ⁺ , Proline, SOD, and CAT as well as K ⁺ , Ca ⁺⁺ , chlorophyll a, and b. This species responds positively to low salt and HM concentrations (100mM NaCl, 50uM NiCl ₂) and moderate salt and HM concentrations (200mM NaCl, 100uM NiCl ₂) while tolerating high-stress levels (400mM NaCl, 200uM NiCl ₂)	(Javeed and Naz 2022)

Table 3 (continued)

Plant species	Heavy metal	Substrate	Treatment	Exposure duration	Effects	References
<i>Trigonella foenum graecum</i>	Cd, Cr, Ni, Al, Pb, As, Se	Nutrient solution	3, 6, 9 mg L ⁻¹	7 days	At 9 mg L ⁻¹ Cd and Pb treatment, root nodules formation was reduced by up to 80%. The fresh nodule's Lb content ranged from 0.4 to 1.8 mg g ⁻¹ , and seven different metals were treated. At 9 mg L ⁻¹ of As treatment, the greatest amount of Lb content generation inhibition was seen in root nodules. In harvested seeds, traces of just two metals, Cd and Ni, were discovered	(Bhandari and Singh 2022)
<i>Satureja hortensis</i>	Cd	Foliar application	0, 75, 100 and 150 µM	At two weeks interval	Cd stress reduced growth indices such as plant height, leaf count, and lateral branch by 25.97%, 19.18%, and 22.39%, respectively. While proline buildup, membrane cell permeability, carbohydrates, and the activity of antioxidant enzymes increased	(Azizi et al. 2021)
<i>Mentha piperita</i>	Cd, Pb, Ni, and As	Solution	Soil with 23.7 mg Kg ⁻¹ As and 5 mg Kg ⁻¹ Cd, soil with 23.7 mg Kg ⁻¹ As, 5 mg Kg ⁻¹ Cd, 136 mg Kg ⁻¹ Ni and 95 mg Kg ⁻¹ Pb	One week	In the first month of exposure, there was no discernible difference between the exposed and the control in terms of growth, development, or chlorophyll content. Effects on the plant were phytotoxic three months after exposure. The fact that the values for the transfer coefficients and translocation factors were typically less than 1 shows that <i>Mentha piperita</i> immobilized the metals in root	(Dinu et al. 2021a)

Table 3 (continued)

Plant species	Heavy metal	Substrate	Treatment	Exposure duration	Effects	References
<i>Calendula officinalis</i>	Ni	Foliar application	0, 0.07, 0.156, 0.234, 0.312 mg L ⁻¹	In two stages: 1st at 2–4 leaves stage and the 2nd at the time of budding stage	Ni concentration of 0.156 mg L ⁻¹ was most effective for improving antioxidant activity and plant development parameters, whereas physiological features were most frequently found at a dosage of 0.234 mg L ⁻¹ . High Ni concentrations showed a negative impact on the indices under investigation	(Sobati et al. 2021)
<i>Salvia sclarea</i>	Cd	Hoagland nutrient medium/ container	0 and 100μM	8 days	Eight days of exposure to Cd, roots and leaves accumulate significantly more Cd than the control, and the root and shoot dry weight per plant is reduced by 36% and 15%, respectively. RWC of leaves is also decreased by 27% in Cd-treated plants on the 8th day of exposure. Lipid peroxidation and oxidative stress disrupt the membrane integrity of leaves due to Cd exposure in such a way that H ₂ O ₂ and malondialdehyde (MDA) in the leaves significantly increased with exposure time by 48 and 42%, respectively, electrolyte leakage is also increased by 12% to 43% in the cell membranes of leaves after exposure for 3 to 8 days. Chlorophyll a and chlorophyll b and carotenoid is decreased by 36%, 32% and 13%, respectively after 8 days. Photochemistry of Photosystem I (PSI) was reduced about 13% in the leaves after 8th day exposure	(Dobrikova et al. 2021)

Table 3 (continued)

Plant species	Heavy metal	Substrate	Treatment	Exposure duration	Effects	References
<i>Lactuca sativa</i>	Cd	Cocopeat and perlite/pot	0, 3, 6 and 9 mM	10 days	For ten days of exposure to Cd, notably decline root and shoot length. Ck plants leaves are flat in appearance whereas plants exposed to Cd have circular yellow leaves. The shoot and root dry weight yields are decreased notably in Cd exposed plants. Plants treated with 9 mM CdCl ₂ decreased total flavonoids (0.05 mg/g FW). The photosynthetic pigmentation is declined in plants treated with Cd	(Kolahi et al. 2020)

HMs Heavy Metals, mg kg^{-1} Milligrams per kilogram, mg L^{-1} Milligrams per litre, mmol L^{-1} , Millimoles per litre, ppm Parts per million, μM Micromolar, mM Millimolar, Se Selenium, Cd Cadmium, Zn Zinc, Cr Chromium, Cu Copper, Ni Nickel, Al Aluminium, Pb Lead, As Arsenic

morphological changes or abnormalities and decrease the amount of bioactive substances in plants. In soil-growing Indian pennywort (*Centella asiatica* L.), applying 50 and 100 mg kg⁻¹ of cadmium nitrate decreased the number of shoots, root length, and centellosides. Centelloside is a bioactive compound with many therapeutic properties (Biswas et al. 2020). The effect of different HMs on medicinal plants' physiological, biochemical, and morphological features are described in Table 3.

4 Biosynthesis of Secondary Metabolites and Its Functions in HMs Stress Tolerance

At low concentrations, certain HMs are essential for plant development and life cycle. These include Zn, which is required for the activity of many enzymes, including SOD, and Ni, a component of the urease enzyme, which is responsible for urea metabolism in plants (Tavanti et al. 2021). These essential HMs and other non-essential HMs, i.e., Cd are toxic to plants and can interfere with their biochemical and morpho-physiological processes when in high concentrations (Noor et al. 2022). Toxic HMs are ecotoxic due to their high bioaccumulation and toxicity, acting as abiotic stressors and damaging plant cells through oxidation. Plants can recognize these threat signals and respond in several defensive ways (Anjitha et al. 2021), which boosts the plant's secondary metabolites (SMs) biosynthesis (Hurmat and Bansal 2020). Metals with an inherent capacity to withstand toxicity can withstand the detrimental effects of toxins on metabolic and physiological functions, allowing them to remain in nature (Sameena and Puthur 2021). Metabolites are classified as primary metabolites, responsible for the main metabolic pathways critical for plant survival, and SMs are only involved in developing the plants' ability to interact with the surrounding adverse environment. Plants restricted to specific taxonomic groupings have almost 100,000 SMs (Ashraf et al. 2018). Plants have different major SMs based on their metabolic processes, such as phenolic chemicals, terpenes, and nitrogen-containing compounds (Ashraf et al. 2018). Plants suffer many physiological changes due to HMs in the soil, and their chemical makeup is also affected (Yin et al. 2022). They also stimulate the biosynthetic pathways, which can cause immunological responses in plants, resulting in SMs buildup (Ashraf et al. 2018), and plants in response, have adopted many strategies at the morphological, anatomical, biochemical, and molecular levels to counteract the impacts of stress (Punetha et al. 2022), which have long-term effects on plant development and survival in harsh conditions (Kurepin et al. 2017). SMs in plants, such as carotenes, flavonoids, and isoprenoids, can confer HMs stress resistance and can be used as tools to deal with varied stressors (Khare et al. 2020) and enhancement of enzymes

involved in the biosynthesis route of phenolic substances such as shikimate dehydrogenase, chalcone synthase, and polyphenol oxidase, phenylalanine ammonia-lyase (Chen et al. 2019b; Zoufan et al. 2020). Phenolics are aromatic SMs with redox characteristics, allowing them to operate as potential antioxidants under stress. A variety of biological macromolecules, including nuclear proteins and DNA, can interact with HMs, causing an increase in ROS such as superoxide anion radicals (O_2^-), hydrogen peroxide (H_2O_2), and hydroxyl radicals (OH^-) (Juan et al. 2021). For instance, the artemisinin content significantly increased when shoot cultures of sweet wormwood (*Artemisia annua* L.) were treated with 5 μM $CuSO_4$. Cu stress causes the production of ROS, which causes dihydroartemisinic acid to become artemisinin (Saghizadeh et al. 2019). In response to 25 μM $CuSO_4$, sweet basil (*Ocimum basilicum* L.) produced more phenylpropanoids, primarily eugenol and methyl eugenol (Trettel et al. 2018). Many studies found that *SpMTL* gene significantly improved plant tolerance to Cd stress, and Cd content in plants was positively linked with *SpMTL* expression (Peng et al. 2017). Cinnamic acid production was increased by adding Co ions to an in vitro growth of the tea plant (*Camellia sinensis* L.) (Sutini et al. 2019). In addition, anthocyanins, a flavonoid with ROS scavenging potential, contribute to increase As tolerance in various plant species. In detoxifying, anthocyanins may combine with As, followed by vacuolar sequestration, similar to glutathione (GSH) and PCs (Ahammed and Yang 2022).

Small RNAs and other regulatory elements have also been implicated in controlling the plant as a stress response (Kumar et al. 2017). Through genetic manipulation of phytochelatin production and glutathione-related genes, SMs' production and function, as well as the activation of the genes responsible for their synthesis, can be changed, increasing plant tolerance and environmental adaptation (Jan et al. 2021). Glutamyl cysteine synthetase (GSH1), glutathione reductase (GR), cystathionine synthetase (CTS), glutathione synthetase (GSH2), glyoxylases, serine acetyltransferase (SAT), ATP sulfurylase (APS) and phytochelatin synthetase (PCS) genes were identified as prospective candidates for controlling overexpression of these genes led in increased tolerance to hazardous metals or metalloids (Shefa et al. 2018; Yadav 2010). Seedlings exposed to Cr showed a significant increase in the expression pattern of the rice (*Oryza sativa* L.) metallothionein genes, *OsMT1b* and *OsMT2c*, upon PCR analysis (Yu et al. 2019a). Metallothioneins are metal-binding proteins that play an important role in metal homeostasis and HMs toxicity protection and play key roles in various physiological processes. Metallothioneins have free radical scavenging activity due to the abundance of cysteine sulfhydryl groups (Dubey et al. 2019; Yu et al. 2019a). For example, the type 4 metallothioneins expressed specifically in *Arabidopsis* seeds, *AtMT4A* and

AtMT4B, perform critical roles in Zn homeostasis. Transcription factors (TFs) are proteins that bind to particular sequences upstream of gene sequences to govern biological growth by changing gene expression (Mitsis et al. 2020), and other HMs like Cd ions induced considerable expression of *BcABCC1* and *BcABCC2* in mustard (*Brassica campestris* L.) var. Chinensis. As a result, increased sequestration of the PC-Cd complex in vacuoles was found, implying that the presence of HMs serves as a stimulant for phytoremediation techniques to attenuate the harmful effects of HMs (Huang et al. 2021), while the identification of essential transcription factors and their roles is an important tool for investigating plant growth and development in response to external abiotic stressors (Shen et al. 2019a). To summarize, some plant species have traits that allow them to absorb, accumulate, and defend against hazardous HMs ions in large quantities (Chaudhry et al. 2020). Therefore, research to understand the mechanisms and regulation of oxidative stress in plants has been prompted by existing knowledge of the genes involved in the response to HMs stress.

5 Response of Medicinal Plants to HMs Toxicity

5.1 Detoxification Role of Non-Enzymatic and Enzymatic Antioxidants in Medicinal Plants

Plants are stationary and unable to escape biotic and abiotic stresses such as HMs toxicity and other range of environmental factors that affect plant growth, development, and output (Yadav et al. 2022). These pressures frequently disrupt plant cell homeostasis and ion distribution, causing osmotic stress and increasing the formation of ROS (Rajput et al. 2021), but some of them can be beneficial at specific stages of cell development, like division and programmed cell death, and they can also act as signature molecules required to regulate gene expression (Huang et al. 2019). Due to ROS's dual behavior, plant cells must adopt methods to balance or limit their production of ROS (Abdelgawad et al. 2020), as antioxidants are the first line of defense against free radical damage and are essential for the optimal health of plant cells (Rajput et al. 2016). Plant antioxidants contribute significantly to plant development through a range of actions and processes. The antioxidant capacity of a plant is the sum of all the activity of all the enzymatic and non-enzymatic antioxidant systems (Sharma et al. 2022).

In plants, several antioxidant enzymes involved in scavenging ROS are present, and oxidative stress is known to increase the synthesis of these enzymes (Hasanuzzaman et al. 2020). Furthermore, oxidative stress is caused by an excess of ROS such as free radicals (superoxide anion, O_2^- ; hydroperoxyl radical, $HO_2^\bullet$; alkoxy radical, $RO^\bullet$; and

hydroxyl radical, $\bullet\text{OH}$) and non-radical molecules (hydrogen peroxide, H_2O_2 , and singlet oxygen, $^1\text{O}_2$) (Hasanuzzaman et al. 2019). Plants primarily deal with oxidative stress via an endogenous defensive mechanism consisting of different enzymatic (catalase, CAT (Nandi et al. 2019); ascorbate peroxidase, APX (Liu et al. 2019); SOD (Stephenie et al. 2020); glutathione reductase, GR (Hasanuzzaman et al. 2017); dehydroascorbate reductase, DHAR (Chang et al. 2017); glutathione peroxidase, GPX (Zhang et al. 2019); monodehydroascorbate reductase, MDHAR (Vanacker et al. 2018); guaiacol peroxidase, GOPX (Pieczul et al. 2020); glutathione S-transferase, GST (Kumar and Trivedi 2018); Ferritin; oxidase-like alternative oxidase, AOX (Wojciechowska et al. 2021); nicotinamide adenine dinucleotide phosphate (NADPH) (Hong et al. 2020); thioredoxins, TRXs (Mata and Spoel 2019); glutaredoxin, GRX; peroxiredoxins, PRXs (Malik et al. 2020)) and non-enzymatic (ascorbic acid, AsA; phenolic acids; glutathione, GSH; alkaloids; carotenoids; α -tocopherol; flavonoids; nonprotein amino acids (Ashraf et al. 2019) antioxidants.

The antioxidant defense mechanism and ROS buildup maintain a steady-state balance in plant cells. The management of various crucial activities for plants, including growth and development, depends on maintaining an ideal ROS level in the cell (Mittler 2017). The harmony between ROS scavenging and production keeps this intermediate level in place (Hasanuzzaman et al. 2019). However, under stressful circumstances, excessive ROS production destroys the equilibrium and results in cellular damage, which lowers plant yield and induces programmed cell death (PCD) (Raja et al. 2017).

A tetrameric heme-containing enzyme called CAT is used to detoxify ROS and can change 26 million H_2O_2 molecules into water in a minute (Mehla et al. 2017). Since CAT is absent in chloroplasts, APX is the only enzyme capable of scavenging H_2O_2 there. AsA-dependent APX occurs in a variety of isoforms, including cytosolic APX (cAPX), mitochondrial APX (mtAPX), chloroplastic APX (chlAPX), peroxisomal/glyoxysomal APX (mAPX), and chloroplastic (MDHA) (Pandey et al. 2017). PhOH is the primary substrate that peroxidase oxidizes to produce the phenoxyl radical ($\text{PhO}\bullet$), also known as QA, in which H_2O_2 absorbs an electron and changes into H_2O . Suberin, lignin, and quinines are created when $\text{PhO}\bullet$ and AsA cross-react in the absence of AsA, whereas MDHA and DHA are created when $\text{PhO}\bullet$ and AsA react (Jovanovic et al. 2018). Non-enzymatically, monodehydroascorbate is reduced to DHA, which is subsequently recycled to AsA via the activity of DHAR that is dependent on GSH (Garcia-Caparrós et al. 2019). This reaction involves the oxidation of GSH to GSSG, which is then reduced back to GSH by the NADPH-dependent GR enzyme, an enzyme that is also essential for maintaining redox homeostasis (Couto et al. 2016).

The AsA-GSH cycle, also known as the Asada–Halliwell cycle, is the primary antioxidant defense route in plant cells for detoxifying H_2O_2 (Akram et al. 2017). The AsA-GSH cycle plays a crucial part in the antioxidant defense system by reducing H_2O_2 and maintaining redox equilibrium (Hasanuzzaman et al. 2019). Additionally, GPX and GST play a crucial role in the detoxification of xenobiotics and H_2O_2 (Hasanuzzaman et al. 2018; Chen et al. 2019a, b, c; Jiang et al. 2023). AsA and GSH are the most prevalent soluble nonenzymatic antioxidants in higher plants. They are essential electron donors and directly scavenge ROS through the AsA-GSH cycle (Hasanuzzaman et al. 2019). Additionally, beta-carotene interacts with $\bullet\text{OH}$, $\text{O}_2\bullet$, and $\text{ROO}\bullet$ radicals to lower their concentration in cells' ROS (Kapoor et al. 2019b).

Ascorbic acid, tocopherol, glutathione, and carotenoids are well-known phenolic compounds among non-enzymatic antioxidants for their ability to bind/chelate HMs and/or scavenge the ROS produced within plant cells (Maleki et al. 2017). Through its ability to donate electrons and maintain stability as a result of electron delocalization brought on by the resonance between two forms, ascorbate plays a crucial role in the AsA-GSH cycle to scavenge ROS (Hasanuzzaman et al. 2019). AsA controls a large number of phytohormone production pathways. Additionally, AsA regenerates α -tocopherol (vitamin E) by scavenging $\bullet\text{OH}$ and $\text{O}_2\bullet$ or from tocopheroxyl radical (Seminario et al. 2017). In severely stressed plants, flavonoids are found in or near areas where ROS are produced, where they serve as antioxidants (Hasanuzzaman et al. 2020). Stress-insensitive and stress-sensitive species, which have weaker initial defenses against ROS and experience more damage from oxidative stress, both experience an increase in flavonoid production (Yu et al. 2019b). Another significant class of antioxidant molecules is carotenoids, which have been shown to scavenge dangerous free radicals and protect light-harvesting complex proteins and the stability of thylakoid membranes (Hussain et al. 2019b). In addition, Hydroxybenzoic and hydroxycinnamic acids, which make up the majority of the antioxidant phenolic acids, exhibit antioxidant action as chelators and scavengers of free radicals, particularly $\text{O}_2\bullet$, $\bullet\text{OH}$, $\text{ROO}\bullet$, and $\text{ONOO}\bullet$ (Parcheta et al. 2021). Alkaloids can also act as antioxidants by scavenging free radicals and preventing H_2O_2 induced oxidation (Hasanuzzaman et al. 2020). Gamma-aminobutyric acid, ornithine, and citrulline, which are not proteins, are also thought to be powerful nonenzymatic antioxidants (Cruzat et al. 2018).

5.2 Role of Nano-Biotechnology in the Elicitation of SMs

Elicitation is a technique for boosting the output of SMs with pharmacological relevance. Plants activate their defensive

mechanism in response to elicitors, which mobilize SMs (Anjum et al. 2019). Plant stress responses occur at both cellular and organismic levels. Receptors on the cell membrane first detect the stress signal. This initiates an intracellular signaling cascade synthesizing secondary signal molecules (Sewelam et al. 2016).

Nano-particles (NPs) operate as innovative and effective elicitors and generate oxidative stress in medicinal plants to increase SMs synthesis. NPs can improve medicinal plants' morphological, physiological, and other biological factors and their production by alleviating biotic and abiotic stressors (Pandita 2022). Although nanotechnology, particularly NPs, has been discovered to alleviate several agricultural-related bottlenecks, with considerable improvements observed in plant growth, nutrient uptake, and plant disease control when compared to conventional systems (Duhan et al. 2017), and can be used as potent abiotic elicitors to start the production of SMs which can increase the expression of genes responsible for producing SMs (Hatami et al. 2019). In-vitro cultures of numerous plant species currently use diverse NPs as creative and successful SM elicitors of SMs (Moharrami et al. 2017; Asl et al. 2019). Carbon (C) nano-tubes, gold (Au), Ag, zinc oxide (ZnO), Cu, and titanium dioxide (TiO₂) NPs are the most often used nano-elicitors. The effect of NPs on SMs production has been observed to be dependent not only on the NPs' features (i.e., the exposure time, the concentration utilized, synthesis origin, and their size) but also on the type of plant culture (Chung et al. 2018). Successful plant SMs elicitation of Ag-NPs has been documented for several plant species and production techniques. In cell suspension systems, Ag-NPs (30 µg L⁻¹) have been reported as an efficient elicitor encouraging the formation of lignans (67.23 mg g⁻¹ DW) (secoisolariciresinol diglucoside and lariciresinol diglucoside) and neolignans (45.9 mg g⁻¹ DW) (dehydrodiconiferyl alcohol guaiacylglycerol-β-coniferyl alcohol ether glucoside) in cell suspension of linseed (*Linum usitatissimum* L.) (Zahir et al. 2019). In Aloe vera (*Aloe barbadensis* miller L.) cell solution treated with Ag-NPs (at 0.625 mg L⁻¹), which is extensively utilized for medical and aesthetic applications, aloin (127%) synthesis was considerably boosted (Burange et al. 2021). St. John's wort (*Hypericum perforatum* L.) cell suspension cultures were treated with 100 ppb iron and zinc nano-oxides, which increased the production of hypericin and hyperforin (Shakya et al. 2019). Linseed cell suspension cultures exposed to Ag-NPs and ZnO-NPs produced much-increased levels of lignans and neolignans quickly (Abbasi et al. 2019). Cell cultures of self-heal (*Prunella vulgaris* L.) stimulated by Au and Ag-NPs produced more polyphenolics. Ag-NPs could enhance the accumulation of biomass and SMs in Pepain *Caralluma tuberculata* L.) callus cells in vitro (Ali et al. 2019). Ag-NPs (4/40 mg/plant) were found to trigger the oxidative defense response in cucumber (*Cucumis*

sativus L.) via increasing phenolics (Zhang et al. 2018). Foliar application of Ag-NPs at 200 ppm to hydroponically produced Rosemary (*Rosmarinus officinalis* L.) for 12 days increased carnosic acid concentration by more than 11%, as did total flavonoids (Soltanabad et al. 2020). In hazel (*Corylus avellana* L.) cell suspension cultures, elicitation of 5 ppm Ag-NPs boosted the formation of the compounds taxanes, taxol, and baccatin III as well as total soluble phenols, but decreased at 10 ppm concentration (Jamshidi and Ghanati 2017). In vanilla (*Vanilla planifolia* L.) Jacks. ex Andrews shoots grown in MS media supplemented with 25 and 50 mg L⁻¹ Ag-NPs (35 ± 15 nm), TPC, and antioxidant capacity were significantly improved but decreased at 100 mg L⁻¹ (Spinoso et al. 2017).

It has also been demonstrated that Cu/CuO NPs are superior inducers of secondary metabolism in plants. Mandarin orange (*Citrus reticulata* L.) plantlets germinated in vitro with CuO-NPs (15–32 nm) and ZnO-NPs (8–32 nm) added separately in a medium at doses of 30 g mL⁻¹ exhibited a considerable rise in total phenolic and flavonoid contents as well as antioxidant capacity (Hussain et al. 2017). Additionally, it was discovered to significantly boost the amount of flavonoid, polyphenol, and tannin in the roots of the Indian medicinal herb Ashwagandha (*Withania somnifera* L.) Dunal (Singh et al. 2018). In addition, CdO-NPs of size 7–60 nm; concentration of 2.03, 0.45, 105 particles cm³ increased the level of ferulic acid and isovitexin in barley (*Hordeum vulgare* L.) plants (Vecerova et al. 2016), while at the application of 10 ppm TiO₂ NP (10–25 nm) boosted plant shoot dry mass and essential oil content in Moldavian Dragonhead (*Dracocephalum moldavica* L.) plants grown under regular watering. It showed increased proline and much decreased H₂O₂ and malondialdehyde levels (Mohammadi et al. 2016). In another study, peppermint (*Mentha piperita* L.) was treated with TiO₂-NPs of 21 nm size at concentrations of 100 and 150 mg L⁻¹ and was documented to significantly increase SMs content by 39.4% and 105.1%, respectively, over control, while boosting menthol output and content by 124.1% and 9.6%, respectively (Ahmad et al. 2018). On foliar application, silicon dioxide (SiO₂) NPs (50 and 100 mg L⁻¹) significantly increased SMs content in peppermint, increasing the menthol content while lowering menthone and menthyl acetate (Ahmad et al. 2020), as well as Methyl jasmonate (MeJA) treatments at various doses (0, 10, 50, and 100 µM), were employed to boost alkaloid output in Capeperiwinkle (*Catharanthus roseus* L.). Plants treated with 75 mg L⁻¹ SNPs (1.4 g/g Fresh Weight) had a high vinblastine concentration. Vinblastine and vincristine levels were highest in plants treated with 50 M MeJA (1.23 and 0.9 g/g FW, respectively) (Shahin 2018). The addition of manganese trioxide (Mn₂O₃) NPs to *Atropa belladonna* (deadly nightshade) shoot tips increased flavonoids, phenolics, and alkaloids in a dose-dependent manner while

activating antioxidant defense enzymes SOD, catalase, ascorbate peroxidase and peroxidase (Tian et al. 2018) while it was discovered that *Stevia rebaudiana* Bert adventitious root cultures that were exposed to bimetallic nano-alloys of Cu and Au at a concentration of 30 mg L⁻¹ increased biomass production, increased the total amount of phenolics and flavonoids, and enhanced antioxidant potential. The most significant impact was seen with AuCu (1:3) NPs (Ghazal et al. 2018). Nano-biotechnology can produce high-value SMs on a large scale in bioreactors for the nutraceutical sectors and pharmaceuticals (Fazal et al. 2019).

6 HMs Pose Health Risks to Humans After Consuming Metal-Toxic Medicinal Plants

Herbal medicine is utilized all over the world. Over one-third of adults in the USA reported using at least one herbal remedy (Rashrash et al. 2017). According to estimates, the market for herbal medicines was worth US \$83 billion in 2019 and

would be US \$550 billion by 2030 (Huck et al. 2022). According to reports, 80% of the world's population relies on herbal remedies for primary healthcare (Uttu et al. 2022). However, because HMs (Pb, Cd, Zn, and Cu) quickly assimilate into the soil through their roots, raw plants, which are the primary components of herbal medicines, are easily polluted (Mukherjee et al. 2022a, b). As a result, they affect human nutrition (Zaynab et al. 2022). In contrast to organic contaminants, HMs are not biodegradable and tend to accumulate in living organisms (Khalef et al. 2022). When levels of essential and non-essential HMs exceed threshold values, genetic, physiological, and morphological issues arise, including growth retardation, malignancy, and increased mortality (Collin et al. 2022), and several cellular organelles and parts of biological systems, including the cell membrane, mitochondria, lysosomes, endoplasmic reticulum, nuclei, and several enzymes involved in metabolism, detoxification, and damage repair, have been documented as a result of the HMs impact (Fu and Xi 2020). Metal ions significantly influence the interactions between the parts of the cell. As a result, HM intake can affect cell cycles

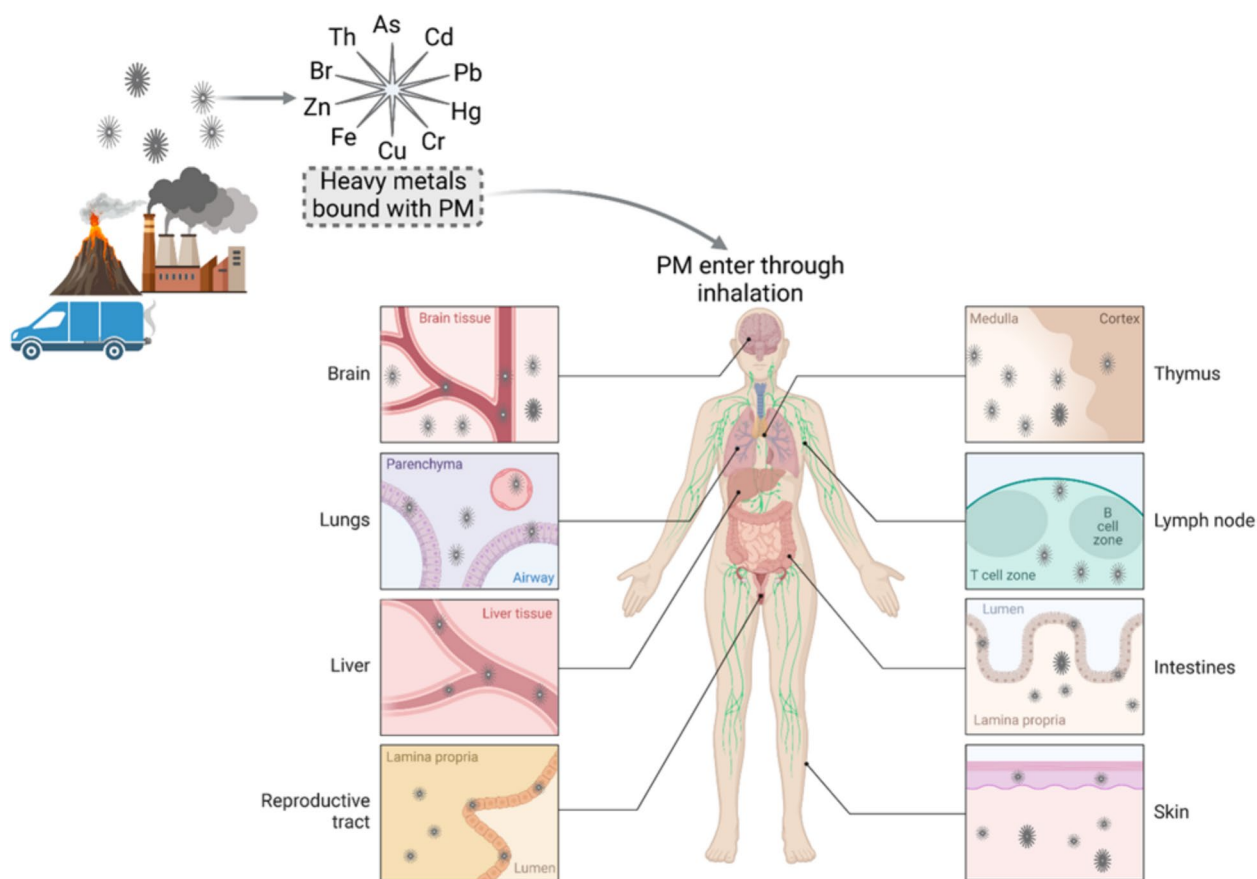


Fig. 2 Effect of Heavy metals on different Human body parts. HMs (Heavy Metals) enters into human body as PM (Particulate Matter) through inhalation and with herbal medicine. HMs effects the nervous, digestive, respiratory and reproductive system of human causing

different diseases. As: Arsenic; Cd: Cadmium; Pb: Lead; Hg: Mercury; Cr: Chromium; Cu: Copper; Fe: Iron; Zn: Zinc; Br: Bromine; Th: Thorium

Table 4 HMs and their impact on human health

Heavy metals	Mechanism of action	Organ affected	Adverse effects	Reference
Pb, Cd, As Hg Ti	By triggering ROS, mitochondrial dysfunction, calcium ion efflux, activation of the immunogenic response, and the inhibition of antioxidants like catalase, SOD, and glutathione,	Brain	HM enter the brain through the blood–brain barrier, causing the emergence of neurodegenerative disorders	(Agnihotri and Kesari 2019)
Cr	Genomic instability, DNA damage, epigenetic modification	Dermal, Kidney, GI	NA	(Pavesi and Moreira 2020)
Cu, Zn, Cd, and Pb	NA	NA	Humans developed anemia in a hypochromic and microcytic form	(Shen et al. 2019b)
Hg	Glutathione peroxide inhibition, enzyme inhibition, Thiol binding	Liver	Particularly in the elderly group (> 60 years), there was a substantial correlation between blood Hg concentrations and the levels of the liver enzymes ALT, AST, and gamma-glutamyl transferase (GGT)	(Chen et al. 2019a)
As	Reduces the amount of Stem Loop Binding Protein (SLBP) in cells, which causes canonical histones (H3.1) to be polyadenylated by default and increases in levels outside of the S-Phase. Polyadenylated H3.1 overexpression and SLBP knockdown both promote malignant cell transformation	carcinogen and has been shown to cause skin, lung, bladder, liver, prostate and kidney cancers	Arsenic-mediated carcinogenesis, including epigenetic alterations, remains largely unknown	(Costa 2019)
Cd	Through its interaction with the MT protein, Cd causes dispositional tolerance. By restricting the distribution of Cd to vulnerable cellular macromolecules, such MT induction safeguards the liver	Liver, Brain, Kidneys	NA	(Lech and Sadlik 2017)
Pb	Increased inflammatory cytokines IL-1 β , TNF- α , and IL-6 in the CNS	Multiple organs: CNS, Liver, heart, lungs	Pb has the ability to disrupt the balance of the oxidant-antioxidant system and trigger inflammatory responses in several organs. Exposure to Pb can also triggers physiological changes in the body	(Kianoush et al. 2012)
As	Thiol binding, damage of capillary endothelium, Inhibition of ATP formation	GI, Liver, hair, skin	The plasma exposed to low levels of arsenic showed higher levels of the proinflammatory cytokines TNF-, IL-6, IL-8, and IL-12. In the exposed population, there was a drop in the level of the anti-inflammatory cytokine IL-10 and also affected airway cells' DNA	(Shen et al. 2013)

Table 4 (continued)

Heavy metals	Mechanism of action	Organ affected	Adverse effects	Reference
Br	NA	Heart	Short-term exposure might result in nausea, diarrhea, abdominal cramps, high or low blood pressure, numbness around the face, and weakness of the muscles. Taking too much Br can result in excessive blood pressure, heart rhythm abnormalities, paralysis, and even death	(Martin and Griswold 2009)
Pb	Binding of thiol, proteins	Brain, CNS, Kidneys	Less Hb counts in blood, hypertension, kidney damage, miscarriages, nervous system disturbances, brain damage, infertility, and intellectual disabilities	(ATSDR 2007)
Cd, Pb	Oxidative stress and ROS, mimic molecules and ions, binding to additional groups besides thiol, MTs Protein, Apoptosis, Changes in cell–cell adhesion, signal transduction changes	Reproductive organ	Asthenozoospermia, teratozoospermia sperm head actin loss In testicular tissue, apoptosis	(Benoff et al. 2000)

HMs Heavy Metals, *GI* Gastrointestinal, *CNS* Central Nervous System, *ATP* Adenosine Triphosphate, *Hb* Hemoglobin, *MTs* Metallothioneins, *ROS* Reactive Oxygen Species, *SOD* Superoxide Dismutase, *ALT* Alanine transaminase, *AST* Aspartate aminotransferase, *TNF* Tumor necrosis factor, *Hb* Hemoglobin, *As* Arsenic, *Cd* Cadmium, *Pb* Lead, *Hg* Mercury, *Ti* Titanium, *Cr* Chromium, *Cu* Copper, *Zn* Zinc, *Br* Bromine

and cause apoptosis, carcinogenesis, and other diseases (Ilieva et al. 2019; Mishra et al. 2022). HMs wastewater is progressively discharged into the environment, primarily in developing countries (Jadeja and Jain 2022) and medicinal plant farms were irrigated with this wastewater. Then people's ingestion of these materials may result in the accumulation of HMs in the body since medicinal plants can absorb harmful metals from the local soil. These metals are found in lower concentrations in the leaves and stems than in the roots. The Health Risk Index (HRI), which compares the metal's concentration in edible portions to its reference dose and the amount taken per unit of body weight, is used to determine the health risk in humans (Ashfaq et al. 2022). To determine the danger of HMs in various medicinal herbs, numerous research has been carried out in a variety of nations, including Pakistan (Hyder et al. 2022; Parvez 2022), Turkey (Karahan 2023), China (Lu et al. 2022), Iran (Keshvari et al. 2021), India (Gyamfi 2021), Saudi Arabia (Ahmad et al. 2022), and Serbia (Krstic et al. 2021). According to reports, most medicinal plants have high factors like roots, shoots, and stems for Cu, Zn, Cd, and Pb accumulation and transfer (Shelar et al. 2021). In another study, the concentrations of Br, Al, Ca, Cr, Cd, Cu, K, Fe, Mn, Mg, Ni, Na, Pb, and Zn in the leaves of 37 medicinal plants in Turkey and the results showed that the necessary precautions must be taken into account to protect the fields where medicinal plants are grown or collected from HMs contamination. Otherwise, it demonstrates that these plants might harm people's health (Ozyigit et al. 2021). Long-term usage of herbal supplements, teas, extracts, or tinctures contaminated with HMs may increase the risk of building up toxic or unsafe levels of HMs in the human body (Cirtina et al. 2022) (Fig. 2). A systematic review was conducted to assess safety in 2022 and concluded that most herbal medications were harmful to consume while being marketed as innocuous, according to a comprehensive evaluation of safety assessments of herbal medicines in terms of toxic metals. There should be sufficient screening from when herbs are collected until they are manufactured into finished herbal medicines and sold (Zamir et al. 2022).

One of the main factors contributing to many of the adverse consequences of HMs is their ability to cross cell membranes (Thakur et al. 2022). The measurement of hazardous metal kinetics is aided by transmembrane transfer since it may be involved in metal absorption, distribution, and excretion due to the significance of cell membranes in metal toxicity (Campbell et al. 2022). Among other toxicities, reproductive toxicity has been linked to HMs like Pb, Ag, Cd, U, V, and Hg, as well as metalloids like Ar, while Pb at moderate to low concentrations may have an impact on several reproductive factors, including semen quality (Mukherjee et al. 2022a, b) in addition, quick entry of Hg into the brain results in tremors, depression, and behavioral problems (Manohar et al. 2021) and other metals like Cd, Ar, and Pb produce free radicals that harm cells by causing

oxidative stress and the oxidation of biomolecules as shown in Fig. 2. Other metals, in contrast, are cancer-causing and neurotoxic (Burnase et al. 2022). Since herbal medications and dietary supplements are not subject to mandatory regulation by international regulatory agencies, these products must be produced by the highest manufacturing standards to guarantee consumer safety (Udom et al. 2022), and the impact of HMs on human health is discussed in Table 4.

7 Conclusion and Future Perspectives

The use of medicinal plants to create dietary supplements or other therapeutic preparations is predicated on the usage of raw materials free of any kind of contamination. Health issues may arise due to the heavy metals (HMs) residues present in medicinal plants, and using such goods increases the possibility of the treatment's combination being unsafe, especially when used for extended periods. HMs are most effectively transported by particulate matter (PM), which transports them into plant cuticles through foliar absorption via stomata. PM is a mixture of both anthropogenic and natural components and a significant source of both required and optional HMs, including Zinc (Zn), Lead (Pb), and Nickle (Ni), and these are dangerous as well as can attach to polysaccharides on the surface of rhizodermal cell walls and are taken up by medicinal plants through their roots. The kind of plant or soil, metal speciation in the soil, soil particle size, cation-exchange capacity, soil pH, organic content, and microbial activity all affect how much HMs are absorbed by roots, while medicinal plants can be classified based on enrichment factors (EF) and translocation factors (TF) which are used to measure how closely a plant is related to the soil in terms of HMs buildup in its shoots and roots. Based on their metabolic process, plants produce major secondary metabolites (SMs), such as phenolic, terpenes, and nitrogen-containing compounds. To stimulate the synthesis of SMs in medicinal plants, Nano-particles (NPs) act as creative and efficient elicitors and cause oxidative stress. By reducing biotic and abiotic stresses, NPs can enhance medicinal plants' morphology, physiological, and other biological characteristics and output. In addition, to maintain a steady-state balance in plant cells, antioxidant defense mechanisms, reactive oxygen species (ROS) accumulation, and other enzymes like catalase (CAT) are all involved, and among non-enzymatic antioxidants, ascorbic acid, tocopherol, glutathione, and carotenoids are well-known phenolic compounds for their capacity to bind/chelate HMs and scavenge the ROS generated within medicinal plant cells. These metals can enter the body by cutaneous contact, ingestion, or inhalation from sources of contamination of these medicinal plants. Once within the body, they are absorbed and bio-accumulated for varying amounts of time in different organs

or target tissues and sources of HMs such as pesticides, fertilizers, mining, and other industries and wastewaters which are being used for irrigation of medicinal plant farms, and they have negative effects on human organs and their activities, including cardiovascular, neurological, gastrointestinal, immunologic, endocrine, and reproductive diseases as well as numerous cancers, including lungs, bladder, and skin cancer. The new breeding techniques and genes should be identified, so medicinal plants can withstand the HMs stress, and rules and regulations should be made at the International and national levels to ensure the HMs free raw materials for the end consumer product.

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Declarations

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